

DAY 7 — THE IRIS CHART STORY: ONE NARRATIVE, FOUR CHARTS

Prepared by: Prasanna Shetty **Dataset:** Iris (150 flowers, 3 species)

1. Dataset Overview

- The dataset tracks four distinct features: sepal length (cm), sepal width (cm), petal length (cm), and petal width (cm).
- The data is evenly divided among three species targets: Iris-setosa, Iris-versicolor, and Iris-virginica.
- Each species is represented by exactly 50 flowers, ensuring a balanced sample size.

2. The Visual Narrative & Chart Breakdown

Read together, the four charts tell one cohesive story about how to accurately identify an iris flower.

- **Sample Composition (Pie Chart):**
 - **Objective:** To answer how the sample is split across species.
 - **Design Choice:** A 3-slice pie chart is utilized to display part-of-whole shares, perfectly satisfying the rule of using 5 slices or fewer.
 - **Takeaway:** Each species supplies exactly 50 flowers (33%), confirming that no single species can dominate or skew the averages.
- **Petal Magnitudes (Clustered Column Chart):**
 - **Objective:** To determine which species has the biggest petals.
 - **Design Choice:** The clustered column format was chosen to compare magnitudes across discrete groups.
 - **Takeaway:** This measurement provides the cleanest split in the data, isolating Setosa's petals and a third of virginica's measurements.
- **Measurement Separation (Line Chart with Markers):**
 - **Objective:** To reveal which specific measurements best separate the species.
 - **Design Choice:** A profile line chart effectively compares the shapes of the 3 series across an ordered measurement axis.
 - **Takeaway:** The species lines separate widely on petal measures but nearly overlap on sepal width, proving that petals carry the primary identifying signal.
- **Sepal Width Pattern (Clustered Bar Chart):**
 - **Objective:** To test if sepal width follows the same distinguishing pattern as petal width.

- **Design Choice:** A horizontal clustered bar displays two metrics side-by-side per group, which allows longer labels to be read more easily.
- **Takeaway:** Setosa introduces a twist: despite having the smallest petals, it has the widest sepals (~3.4 cm), demonstrating that a classification rule based on sepal width alone would point to the wrong flower.

3. Final Conclusion

- The primary finding is that petal measurements are the reliable identifiers within this dataset.
- Any successful classification rule must weight petal measurements above sepal width to avoid misidentification.